BEHIND THE SETS

lEgoarCh

Name a building → get a buildable LEGO set.

The engineering under the demo → how a sentence becomes real, orderable bricks.

Generative AI · Emilie El Chidiac & Charles Abi Chahine · 2026

Computation you can hold.

Three reasons a generated set beats a render.

Real, not rendered

Generative form that snaps together in real bricks.

FOR THE COMPUTATIONAL DESIGNER

Models, not mockups

Hand a client a set they keep → not foam they bin.

FOR THE PRACTICING ARCHITECT

Iterate in bricks

Change the design → reorder only what changed.

FOR THE STUDENT & MAKER

A form a machine dreamed up → that real bricks can actually build.

One pipeline, end to end.

Five moves turn a sentence into an orderable set.

Prompt
a building name

→

FLUX.2 + LoRA
→ a set-look render

→

TRELLIS.2
→ a 3D mesh

→

Legolize
→ real bricks

→

Catalog
→ orderable parts

GENERATIVE AI · proposes the form

DETERMINISTIC CODE · proves & orders

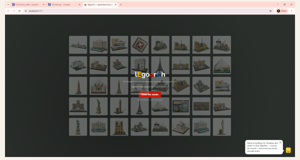
Anyone can generate a picture → the back half, the code that proves it builds, is the work.

THE EXPERIENCE

What you actually do.

One sentence in → a real set on your shelf. Five moves, every one yours to redo.

1 Name it



type any building

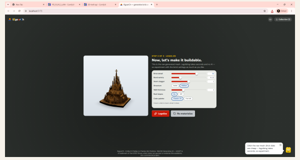
→

2 Forge it

FLUX render · tune it

→

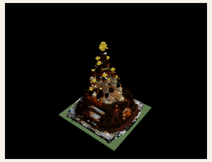
3 Carve it



TRELLIS 3D mesh

→

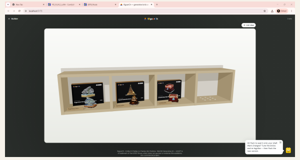
4 Build it



course by course

→

5 Keep it



box · booklet · parts · shelf

Every stop is editable → re-roll the render, re-tune the bricks, reopen any set from the shelf.

One dreams → one carves.

Two AI models do all the imagining. Everything after is plain code.

FLUX.2 Klein 4B

base image model → dreams the picture

+ legoarch LoRA · 40 real Architecture sets

→

TRELLIS.2-4B

render → carves a textured 3D mesh

invents the faces the photo can't see

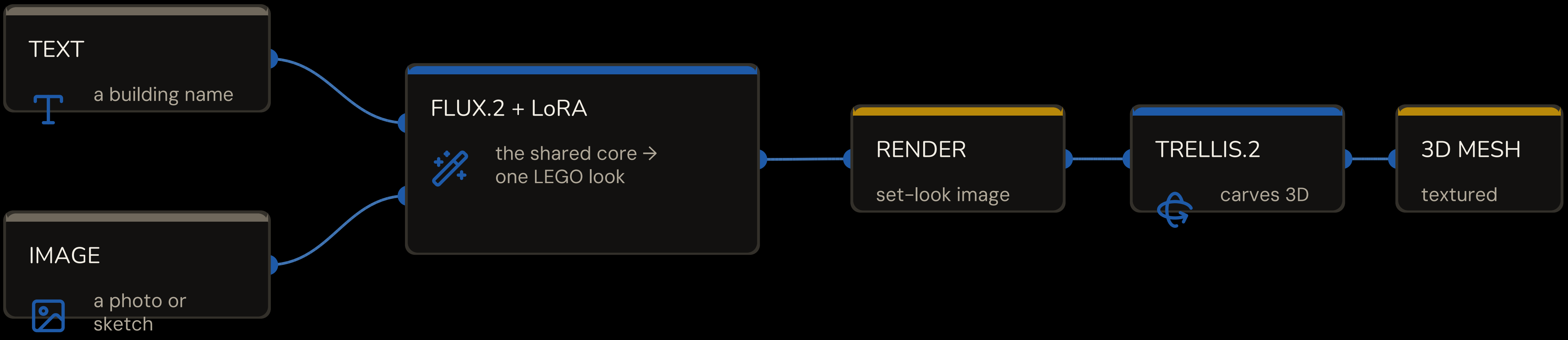
qwen3-4b encoder · flux2-vae · euler · 1024×1024

swept, then fixed → 28 steps · CFG 5.0 · LoRA 1.0 · negative prompt on

Two models dream → everything after them is plain, checkable code.

Three graphs, one core.

Text or image in → the same FLUX.2 + legoarch LoRA core → then TRELLIS to 3D.



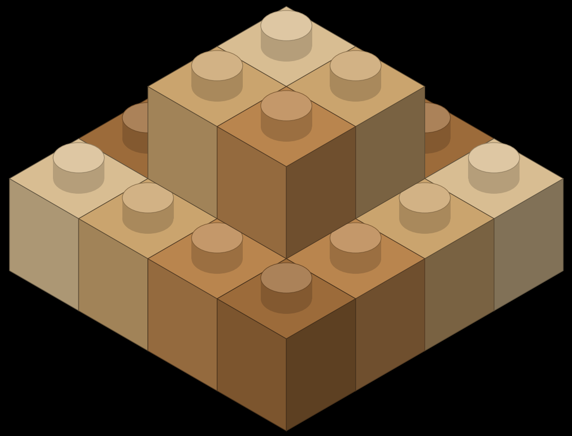
txt → img
05_FLUX.2_LoRA.json

img → img
FLUX.2_image-to-image_LoRA.json

img → 3D
3D.json

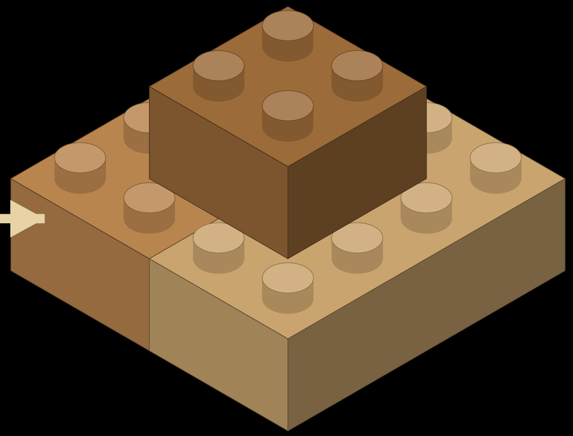
Many 1×1s → a few real parts.

Split-and-merge packs the voxel grid into the largest legal bricks — fewer parts, same shape.



raw voxel grid

thousands of 1×1 cells



merged real parts

2×4 · 2×2 · slopes

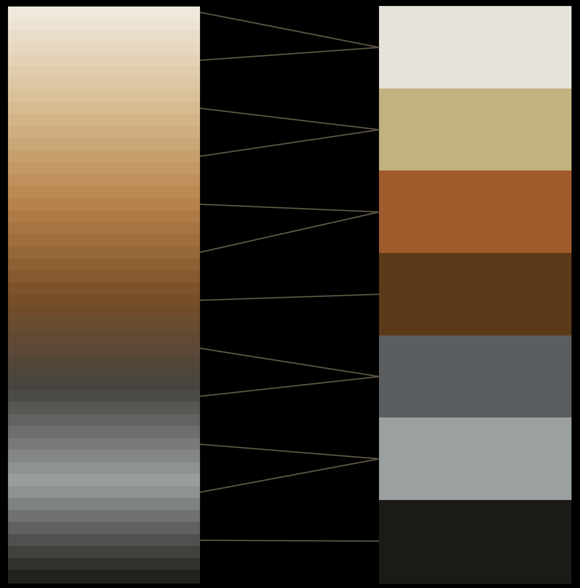
FOUR PASSES

- slopes
- bricks
- plates
- tiles

Every brick is colour-uniform and seam-staggered → every footprint is one a real mould was made in.

Every voxel → a real colour.

A continuous mesh colour is snapped to the nearest of 48 real LEGO colours by ΔE2000.

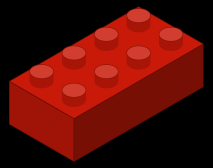


1. **Sample**
   read each voxel's colour off the textured mesh
2. **Match**
   rescale exposure to the FLUX render (per-channel quantile)
3. **Convert**
   sRGB → CIE Lab (D65)
4. **Snap**
   nearest of 48 real colours by CIEDE2000
5. **Clamp**
   to a colour that part was really moulded in

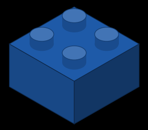
Continuous mesh colour → a small set of real, orderable LEGO colours. No invented paint.

Real parts. Real moulds.

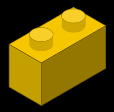
A small vocabulary of actual BrickLink parts → snapped to real colours.



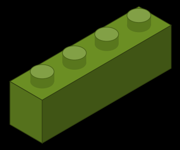
2×4
3001



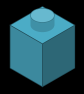
2×2
3003



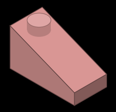
1×2
3004



1×4
3010



1×1
3005



slope
3040

48 real LEGO colours

44
real parts

1,598
validated combos

WE TESTED THIS

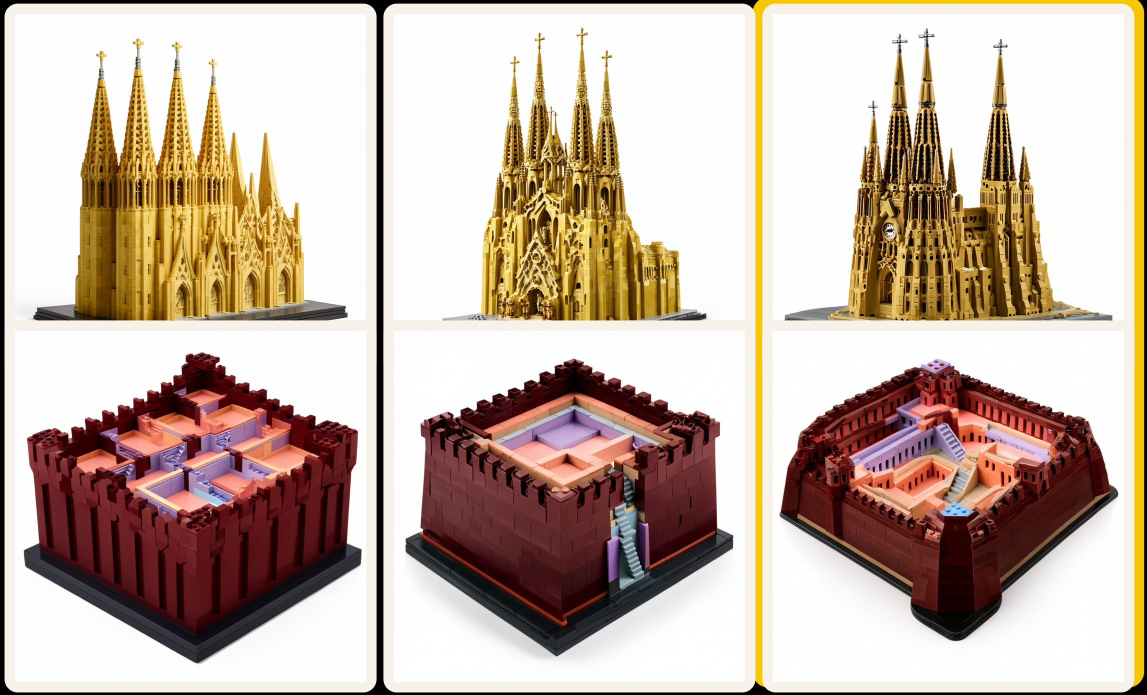
The look lives in the LoRA.

Same prompt, same seed → only the LoRA strength changes.

LoRA 0 | LoRA 0.5 | LoRA 1.0

Sagrada

La Muralla



winner

0 → 0.5 → 1.0

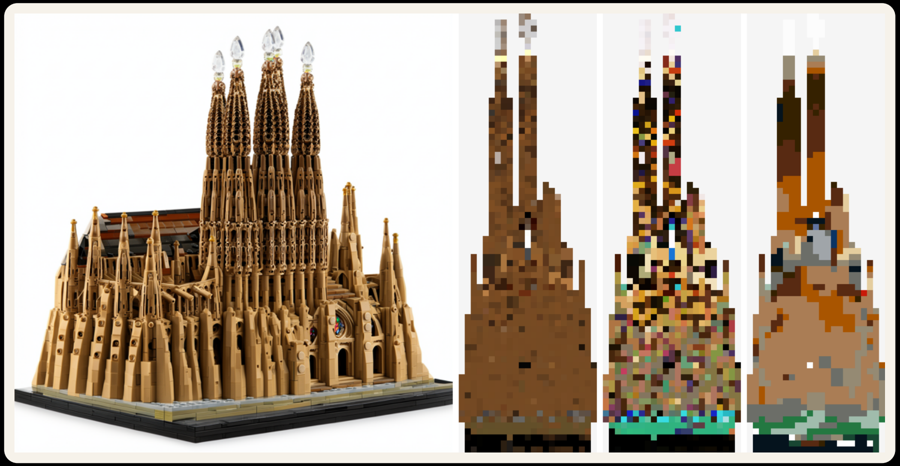
At 0 it's a plain building. At 1 the studs and seams snap in → on a landmark and an obscure block alike.

The LEGO look isn't the prompt. It's the fine-tune.

swept 0 → 1.5 · winner 1.0

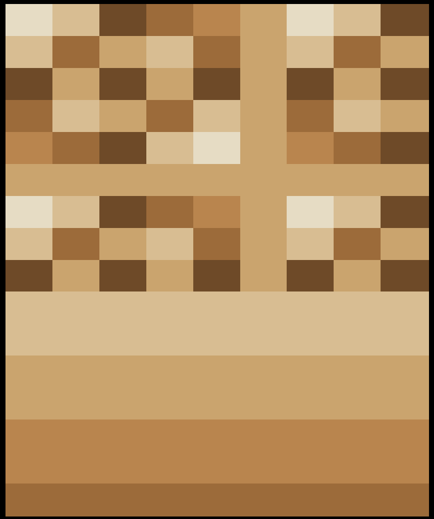
Same building, −46% bricks.

Raw mesh colour speckles into thousands of 1×1s → we merge it to the true colour first.



real build → raw colour grid → merged to true colour

1×1 speckle



merged regions

SAGRADA FAMÍLIA

−46%

8,627 → 4,682 pieces

WHY FEWER IS BETTER

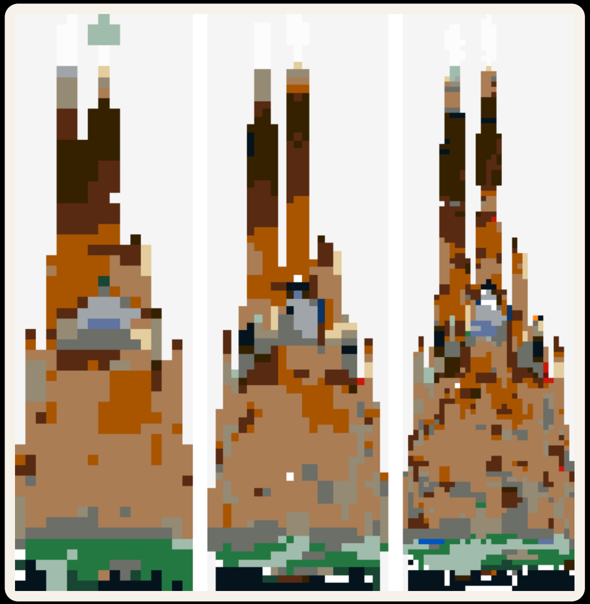
cheaper · simpler to build · same stability

holds across buildings → −19% to −46%

WE TESTED THIS

Draft to flagship — same set.

Pick the detail; the method holds. Pieces grow about quadratically.



Sagrada Família at detail 24 · 32 · 48

detail 24
draft
2,189 pieces

detail 32
default · winner
4,682 pieces

detail 48
large
15,089 pieces

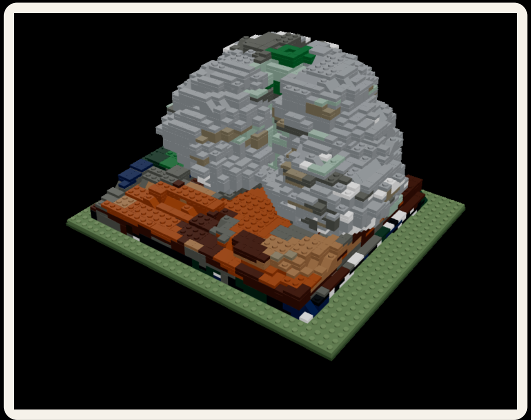
16–64 studs available in-app · same connected set every time

It built Gehry, not the Guggenheim.

We named both. The model followed the curves — and voxels can't hold them.

THE PROMPT — VERBATIM

"Guggenheim Museum Bilbao Frank Gehry … interconnected swirling titanium-clad volumes … overlapping curved ship-like masses merging into one continuous sculptural body …"



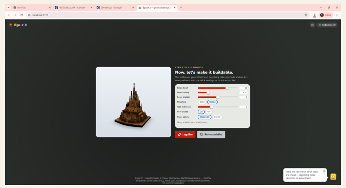
voxelized Bilbao → a connected metallic blob

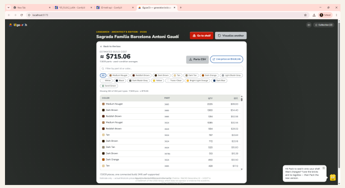
Smooth curves can't survive a voxel grid.

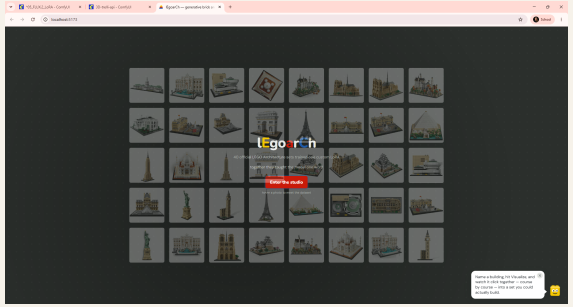
Connected and 93% supported → just not legible. Knowing exactly where the method ends is part of the work.

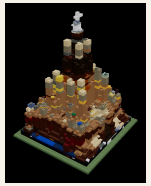
One sentence → everything you get.

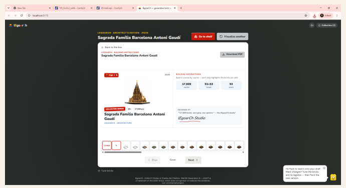

the render

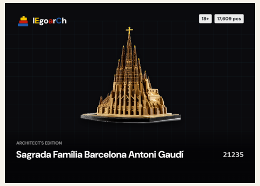

the 3D mesh


parts + price


NAME A BUILDING


the build


the booklet



the boxed set

lEgoarCh

From words to LEGO.

Generative AI proposes → code proves → the catalog ships.

Emilie El Chidiac & Charles Abi Chahine · 2026